A Comprehensive Literature Survey on Detecting Parkinson's Disease Using Vocal Biomarkers Based on Speech Foundation Models

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1 Introduction and Research Context

1.1 Background

Parkinson's Disease (PD) is a progressive neurodegenerative disorder primarily characterized by motor symptoms including tremors, rigidity, and bradykinesia. According to recent epidemiological data:

Table 1: Epidemiological statistics related to Parkinson's Disease.

Statistic	**Value**
Annual new cases in US	$\sim$ 900,000
Current US prevalence	$\sim$ 1 million
Projected US prevalence by 2023	$\sim$ 1.2 million
Global ranking	2nd most common neurodegenerative disease

The Diagnostic Challenge:

- **21%** of patients report visiting their general practitioner three or more times before specialist referral.
- **70–90%** of PD patients experience speech disorders (hypokinetic dysarthria).
- **Prodromal phase**: Emerging symptoms can occur up to **5 years** before significant motor impairments.
- **Speech impairments are pervasive** and offer a non-invasive, cost-effective biomarker.

1.2 The Role of Speech Foundation Models

What are Speech Foundation Models?

Speech foundation models are large-scale neural networks pre-trained on vast amounts of audio data in a self-supervised or weakly supervised manner. They offer several advantages for clinical speech processing:

1.3 Previous Approaches and Their Limitations

Conventional Approaches:

Key Limitations of Existing Work:

Table 2: Advantages of speech foundation models for clinical speech processing.

Advantage	Description
Generalization	Transferable across different tasks and domains
Multi-linguality	Can process multiple languages without retraining
Robustness	More resilient to noise and recording variations
Efficiency	Effective in low-resource settings
Representation Learning	Automatically extract task-relevant features

Table 3: Evolution of approaches for speech-based Parkinson's Disease detection.

Era	Approach	Key Studies	Limitations
Classical ML	Handcrafted features + SVM/RF	Liu et al. (2023), Botelho et al. (2024)	Language-specific, fragile to recording conditions
Supervised DL	End-to-end CNNs, LSTMs	Faragó et al. (2023), Vásquez-Correa et al. (2018)	Data-hungry, overfitting in small datasets
Foundation Models	Wav2Vec 2.0, HuBERT, WavLM	La Quatra et al. (2024), Favaro et al. (2023)	Emerging but need systematic evaluation

1. **Constrained Speech**: Many studies rely on specific vocalization tests (e.g., /a/ phonation) or read speech, which is inconvenient in real clinical settings.

2. **Fine-tuning Challenges**: Fine-tuning foundation models is difficult with small datasets.

3. **Language-Specific**: Most studies focus on single languages (Spanish, English, Czech).

4. **Task-Specific**: Models optimized for specific tasks may not generalize.

1.4 Research Objectives

Primary Objectives:

1. Introduce a new US English PD dataset collected in real-life clinical environments with clinical neurologist labels.

2. Provide comparative results across a substantial number of off-the-shelf speech foundation model variants (without fine-tuning).

3. Assess models on unconstrained, free, and open-vocabulary speech.

4. Use efficient, shallow downstream classifiers to avoid overfitting in small-data regimes.

Key Innovation: Focus on **unconstrained conversational speech** rather than controlled vocalization tasks, making the approach more practical for real-world screening and doctor-patient interactions.

2 Methodology

2.1 New US English PD Dataset

Data Collection Sites:

Site	Location	Type
Beth Israel Deaconess Medical Center	Boston, MA, USA	Clinical research
University of Massachusetts Chan Medical School	Worcester, MA, USA	Academic medical center

Inclusion Criteria:

Group	Criteria
PD	Confirmed diagnostic status, age ≥ 18
PD Exclusions	Developmental or acquired dysarthria unrelated to PD
HC	Age ≥ 18, free from PD or other neurodegenerative disorders

Participant Demographics:

Category	N	Female	Male	Age (mean $\pm$ SD)
PD	81	24	57	68.2 ± 8.8
HC	81	43	38	40.8 ± 15.0
Total	**162**	**67**	**95**	54.5 ± 18.5

Gender Distribution:

- PD: 29.6% Female, 70.4% Male.
- HC: 53.1% Female, 46.9% Male.
- **Total**: 41.4% Female, 58.6% Male.

Task Type	Tasks	Description
Unconstrained	Cookie Theft Picture	Descriptive analysis
Unconstrained	Procedural Narrative	"Tell me how to make a sandwich"
Unconstrained (Focused)	Open-ended Question	"How are you feeling today?"
Constrained	Paragraph Reading	The Caterpillar Passage
Constrained	Stroop Color Word Test	All components

Age Disparity: The PD group is **27.4 years older** on average than the HC group.

Speech Protocol (6-minute tablet-based):

Task Selection Rationale:

Task	Rationale
Caterpillar Passage	Optimized for movement disorders; increased complexity; greater phoneme coverage; varied utterance lengths; lower reading level; increased prosodic variation
Stroop Test	Sensitivity to early PD deficits; assesses lexical processing, articulatory speed, and interference susceptibility

Recording Setup:

Parameter	Specification
Device	Samsung Galaxy Tab S8
Position	Stand, ∼12 inches from mouth
Microphone	Internal tablet microphone
Sampling Rate	44.1 kHz (downsampled to 16 kHz)
Bit Depth	16-bit
Environment	Quiet room, quiet environment
Application	Proprietary guided protocol

Focused Task Analysis:

The study focused specifically on the open-ended question **"How are you feeling today?"** to assess unconstrained, free speech in a real-world clinical context.

Audio Duration Statistics:

2.2 Data Organization

Cross-Validation Strategy:

Category	Mean Duration ± SD	Min-Max Range
PD	37.6 ± 12.0 s	[12.2, 66.5]
HC	39.9 ± 16.5 s	[2.2, 80.4]

Parameter	Value
Approach	Stratified k-fold CV
k	5
Stratification	Category (PD/HC) AND Gender
Split Sizes	60% Train, 20% Validation, 20% Test
Subject Exclusivity	Each subject appears in only one split

Justification: Stratified by both label and gender to prevent sampling bias and ensure balanced representation across folds.

2.3 Audio Pre-processing

Step	Tool	Purpose
Voice Activity Detection	Silero VAD v5 (default parameters)	Remove non-speech parts (background noise, pauses)
Duration Filtering	None applied	Focus on voice characteristics only

Rationale: The study focused on vocal biomarkers representing only voice characteristics, excluding information such as pause lengths, frequency, or background noise. This approach ensures features reflect vocal quality rather than speech timing patterns.

2.4 Features: Speech Foundation Model Embeddings

Embedding Generation Process:

1. Raw audio input → Foundation Model.
2. Last hidden layer activations extracted (prior to tokenization).
3. Frame-level embeddings → **Averaged across time**.
4. **Suprasegmental feature vector** generated (single representation per audio).

Four Foundation Model Families Evaluated:

2.4.1 Family 1: Wav2Vec2

Model Variant	Parameters	Embedding Dim	Key Feature
Base	94 M	512	Earliest SSL audio foundation model
Base 960h	94 M	512	Fine-tuned on 960h LibriSpeech
Large 960h	315 M	512	Large version, fine-tuned
XLS-R 300m	315 M	512	Cross-lingual, 300M params
XLS-R 2b	2.1 B	512	Largest, cross-lingual

Wav2Vec2 Background:

- Proposed by Baevski et al. (2020).
- Transformer-based framework.
- Self-supervised learning.
- Contrastive learning objective: distinguish correct vs. incorrect latent representations.
- One of the earliest audio foundation models.

2.4.2 Family 2: WavLM

Model Variant	Parameters	Embedding Dim	Key Feature
Base	94 M	512	Enhanced Wav2Vec2
Large	315 M	512	Large version

WavLM Background:

- Builds upon Wav2Vec2 with improvements.
- Enhanced performance in noisy environments.
- Better robustness to acoustic variations.

2.4.3 Family 3: HuBERT

HuBERT Background:

- Proposed by Hsu et al. (2021).

Model Variant	Parameters	Embedding Dim	Key Feature
Base 1s960	94 M	768	Self-supervised learning via masked prediction
Large 1160k	315 M	1024	Larger version, higher dimensionality

- Hidden-unit BERT.
- Self-supervised via masked prediction.
- Builds on BERT concepts adapted for audio.

2.4.4 Family 4: Trillsson

Model Variant	Architecture	Parameters	Embedding Dim	Key Feature
v1	ResNet	5.5 M	1024	Smallest, distilled from CAP12
v2	EffNetv2	8.1 M	1024	EfficientNet architecture
v3	EffNetv2	21.5 M	1024	Larger EffNetv2

Trillsson Background:

- Proposed by Shor & Venugopalan (2022).
- Knowledge distillation from CAP12 model.
- Small and efficient networks.
- Universal paralinguistic speech representations.
- Previously used in healthcare domains (anxiety, depression, MCI).

2.5 Machine Learning Models

Rationale for Shallow Classifiers:

Classifier Architectures Evaluated:

FF-DNN Architecture:

```
Input (Embedding Dimension D)
  |
```

Challenge	Solution
Small dataset (162 subjects)	Use shallow models with few trainable parameters
Avoid overfitting	Simple architectures, regularization
Foundation model embeddings	Already contain rich features; simple classifiers sufficient
Computational efficiency	Fast training and inference

Classifier	Parameters	Key Settings
Support Vector Machine (SVM)	RBF kernel	$C = 10$
Logistic Regression	L2 regularization	LBFGS optimizer
Random Forest (RF)	100 estimators	Gini impurity, sqrt(features) max features
XGBoost	sklearn defaults	Gradient boosting
Feed-Forward DNN	2 layers	320 $\rightarrow$ 160 nodes $\rightarrow$ softmax

```
  v
Dense Layer 1 (320 nodes, activation = ReLU)
  |
  v
Dense Layer 2 (160 nodes, activation = ReLU)
  |
  v
Softmax (2 classes: PD/HC)
```

Optimization:

- **Learning rate**: Initial 1×10^{-3}.
- **Decay rate**: 0.1 (inverse time decay).
- **Optimizer**: Adam.

2.6 Evaluation Metrics

Note: UAR $\approx$ Standard Accuracy in this study due to the balanced dataset (81 PD, 81 HC).

Metric	Definition	Clinical Relevance
Sensitivity	$\frac{TP}{TP+FN}$	Ability to detect PD cases (avoid false negatives)
Specificity	$\frac{TN}{TN+FP}$	Ability to identify HC cases (avoid false positives)
Unweighted Average Recall (UAR)	$\frac{\text{Sensitivity}+\text{Specificity}}{2}$	Balanced metric, insensitive to class imbalance
Area Under Curve (AUC)	ROC curve integral	Threshold-independent discrimination

3 Results

3.1 Foundation Model Comparison

Results with Random Forest Classifier:

Table 4: Performance comparison of foundation model variants using Random Forest.

Model Family	Variant	D	Parameters	UAR	Sensitivity	Specificity	AUC
Trillsson	v1	1024	5.5 M	0.94	0.94	0.94	0.98
Trillsson	v2	1024	8.1 M	0.94	0.95	0.93	0.98
Trillsson	v3	1024	21.5 M	0.94	0.96	0.91	0.97
HuBERT	Base 1s960	768	94 M	0.93	0.94	0.93	0.97
HuBERT	Large 1160k	1024	315 M	0.97	0.98	0.96	0.97
Wav2Vec2	Base	512	94 M	0.86	0.86	0.86	0.94
Wav2Vec2	Base 960h	512	94 M	0.86	0.86	0.86	0.94
Wav2Vec2	Large 960h	512	315 M	0.87	0.88	0.86	0.94
Wav2Vec2	XLS-R 300m	512	315 M	0.88	0.86	0.89	0.93
Wav2Vec2	XLS-R 2b	512	2.1 B	0.86	0.86	0.85	0.91
WavLM	Base	512	94 M	0.80	0.73	0.88	0.86
WavLM	Large	512	315 M	0.87	0.85	0.89	0.93

Key Observations:

1. **HuBERT Large 1160k achieves the highest UAR (0.97)** and highest sensitivity (0.98).

2. **Trillsson v1 achieves the highest AUC (0.98)** with only 5.5M parameters (efficient).

3. **WavLM Base performs worst overall** (UAR 0.80, AUC 0.86).

4. **Wav2Vec2 models show moderate performance** (0.86–0.88 UAR).

5. **Larger not always better**: XLS-R 2b (2.1B) underperforms smaller models.

Performance Ranking:

Rank	Model	UAR	Reason
1	HuBERT Large 1160k	0.97	Best overall accuracy
2	Trillsson (all versions)	0.94	Best efficiency-accuracy tradeoff
3	HuBERT Base	0.93	Good performance
4	Wav2Vec2 XLS-R 300m	0.88	Moderate
5	Wav2Vec2 (others)	0.86–0.87	Moderate
6	WavLM Large	0.87	Moderate
7	WavLM Base	0.80	Worst performance

3.2 Classifier Comparison

Comparison using HuBERT Large 1160k features:

Classifier	UAR	Sensitivity	Specificity	AUC
Random Forest	0.97	0.98	0.96	0.97
XGBoost	0.96	0.96	0.96	0.97
SVM (RBF)	0.96	0.96	0.96	0.97
Logistic Regression	0.94	0.95	0.93	0.96
FF-DNN (2-layer)	0.94	0.93	0.96	0.96

Key Observations:

1. **Random Forest performs best** across all metrics.

2. **Simple classifiers (RF, XGBoost) outperform neural networks** on foundation model embeddings.

3. **All classifiers achieve UAR** ≥ 0.94, showing robustness of HuBERT embeddings.

4. **AUC** ≥ 0.96 **for all classifiers**, confirming strong discriminative power.

Finding	Detail	Implication
Best Foundation Model	HuBERT Large 1160k	UAR 0.97, Sensitivity 0.98
Best Efficient Model	Trillsson v1	AUC 0.98, only 5.5M parameters
Best Classifier	Random Forest	UAR 0.97 with HuBERT
Architecture Insight	Larger not always better	XLS-R 2b underperformed
Model Ranking	HuBERT > Trillsson > Wav2Vec2 > WavLM	Clear hierarchy
Key Innovation	Unconstrained speech works	Real-world feasibility confirmed

3.3 Key Findings Summary

4 Critical Analysis

4.1 Strengths

1. **New Clinically-Annotated Dataset:**
   - 162 subjects (81 PD, 81 HC).
   - Clinical neurologist labels.
   - US English (fills geographic gap).
   - Real-life clinical environment.
2. **Comprehensive Model Comparison:**
   - 12 foundation model variants.
   - 5 classifier architectures.
   - Systematic evaluation.
3. **Focus on Unconstrained Speech:**
   - "How are you feeling today?" open-ended question.
   - More practical for real-world screening.
   - Closer to clinical conversation.
4. **No Fine-tuning Required:**
   - Off-the-shelf models.
   - Avoids overfitting in small-data regime.
   - More practical for deployment.

5. **Efficient Classifiers:**
    - Random Forest performs best.
    - Simple, interpretable, fast.
6. **Balanced Dataset:**
    - 81 PD, 81 HC.
    - Stratified by gender.
    - Reduces class imbalance issues.

4.2 Limitations

Critical Limitations:

Limitation	Details	Impact
Age Mismatch	PD: 68.2 ± 8.8, HC: 40.8 ± 15.0	27.4 year difference; potential confound
Gender Imbalance	PD: 70.4% male, HC: 46.9% male	Dataset bias
Single Task	Only "How are you feeling today?"	May not capture full speech impairment
Single Language	US English only	Limited cross-lingual generalization
No External Validation	Only internal 5-fold CV	Generalizability unknown
Commercial Affiliation	Canary Speech Inc.	Potential bias
No Comparison with Supervised Methods	Only zero-shot/foundation models	Baseline comparison missing

4.3 Comparison with Previous Work

Key Difference: The current study achieves **higher performance (UAR 0.97)** on **unconstrained speech** compared to previous studies, which mostly used controlled tasks.

4.4 Model Architecture Insights

Why HuBERT Large Works Best:

Study	Approach	Dataset	Performance
Current Study	HuBERT + RF	US English (162 subjects)	**UAR 0.97**
Favaro et al. (2023)	SSL features vs. handcrafted	5 languages	F1 0.85 (mono), 0.79 (cross)
La Quatra et al. (2024)	WavLM Base	PC-GITA	AUC 0.82, Acc 0.82
Purohit et al. (2025)	Whisper + adapters	PC-GITA	Acc 0.85
Liu et al. (2023)	Handcrafted features	Multiple	F1 up to 0.85

Factor	Explanation
Self-Supervised Learning	Masked prediction of hidden units captures rich representations
Large Capacity	315M parameters, 1024-dimensional embeddings
Vast Pre-training	Trained on diverse audio data
Architecture	Transformer-based, captures long-range dependencies

Why Trillsson Works Well:

Factor	Explanation
Knowledge Distillation	Learned from larger CAP12 model
Paralinguistic Focus	Designed for speech characteristics
Healthcare Domain	Previously validated on anxiety, depression, MCI
Efficient	5.5M parameters, 1024-dimensional embeddings

Why Larger Models May Underperform:

5 Broader Context and Related Work

5.1 Foundation Models in PD Detection

Evolution of Foundation Model Applications:

5.2 Unconstrained Speech in PD Detection

Shift from Controlled to Unconstrained Speech:

Clinical Value of Unconstrained Speech:

- More representative of real communication.

Factor	Explanation
Overparameterization	2.1B parameters may overfit to pre-training domain
Domain Shift	Clinical speech differs from general audio
Embedding Quality	512-dimensional embeddings may be too compressed
Architecture Differences	Not all models are equally suitable for clinical speech

Year	Study	Approach	Key Finding
2020	Baevski et al.	Wav2Vec 2.0	SSL speech representations
2021	Hsu et al.	HuBERT	Masked prediction of hidden units
2023	Favaro et al.	SSL vs. handcrafted	Embeddings outperform features in cross-lingual scenarios
2024	La Quatra et al.	WavLM + enhancement	AUC 0.82 on PC-GITA
2025	Purohit et al.	Layer selection vs. adaptation	85% accuracy with Whisper + adapters
2025	**Current Study**	**HuBERT + RF**	**UAR 0.97 on unconstrained US English**

Speech Type	Advantage	Disadvantage
Controlled (Vowel, DDK)	Clean signal, consistent, sensitive to specific impairments	Not natural, less practical
Read Speech	Standardized, reproducible	Requires literacy, reading ability
Spontaneous Speech	Natural, clinically relevant, rich features	Variable duration, background noise

- Captures prosody, articulation, and fluency simultaneously.
- Can be collected passively during clinical visits.
- Better for longitudinal monitoring.

5.3 Self-Supervised Learning in Healthcare

Application	SSL Model	Performance
PD Detection	HuBERT, Wav2Vec 2.0	UAR 0.97 (current)
Anxiety Detection	Trillsson	Good performance
Depression Detection	Trillsson	Good performance
MCI Detection	Trillsson	Good performance
Cognitive Impairment	Wav2Vec 2.0	Under investigation

5.4 Efficiency-Accuracy Tradeoff

Model Size vs. Performance:

Model	Parameters	UAR	Efficiency
Trillsson v1	5.5 M	0.94	**Most efficient**
HuBERT Base	94 M	0.93	Moderate
HuBERT Large	315 M	0.97	**Best performance**
XLS-R 2b	2.1 B	0.86	Underperforms

Practical Implications:

- **For resource-constrained devices**: Trillsson v1 (5.5M, AUC 0.98).
- **For maximum accuracy**: HuBERT Large (315M, UAR 0.97).
- **Mobile deployment**: Trillsson feasible (distilled models).
- **Cloud/Server**: HuBERT Large optimal.

6 Implications and Future Directions

6.1 Clinical Implications

1. **Real-World Screening Feasibility:**

- Unconstrained speech ("How are you feeling today?") is sufficient.
- No need for specific vocalization tests.
- Can be integrated into clinical conversations.

2. **Non-Invasive, Cost-Effective:**
   - Tablet-based recording.
   - Automatic analysis.
   - Scalable deployment.
3. **Potential for At-Home Monitoring:**
   - Simple smartphone/tablet setup.
   - Remote assessment.
   - Longitudinal tracking.
4. **Healthcare System Benefits:**
   - Earlier identification.
   - Specialist referral optimization.
   - Resource allocation.

6.2 Technical Implications

1. **Foundation Models are Effective:**
   - HuBERT Large best performance.
   - Trillsson best efficiency.
   - No fine-tuning required.
2. **Simple Classifiers Work Well:**
   - Random Forest matches or exceeds neural networks.
   - Less overfitting.
   - More interpretable.
3. **Unconstrained Speech is Viable:**
   - Captures meaningful PD signals.
   - More practical than controlled tasks.
   - Opens new deployment possibilities.

Direction	Rationale	Priority
Multi-language Validation	Test generalization across languages	High
External Validation	Independent dataset testing	Critical
Longitudinal Studies	Track disease progression over time	High
Age/Gender Controls	Address demographic biases	High
Fine-tuning Exploration	Assess adaptation benefits	Medium
Interpretability	Clinical adoption requires explainability	High
Multi-task Learning	Severity estimation, staging	Medium
Mobile Deployment	Real-world app development	High
Prospective Validation	Clinical trial testing	Critical

6.3 Future Research Directions

6.4 Addressing Current Limitations

Limitation	Proposed Solution
Age Mismatch	Age-matched controls in future studies
Gender Imbalance	Balanced recruitment, subgroup analysis
Single Task	Include multiple unconstrained tasks
Single Language	Multi-lingual dataset collection
No External Validation	Cross-dataset testing
Commercial Affiliation	Independent validation studies
No Supervised Comparison	Include conventional ML baselines

7 Conclusion

7.1 Summary of Key Findings

1. **Best Foundation Model**: HuBERT Large 1160k achieves **UAR 0.97** and **Sensitivity 0.98**.

2. **Best Efficient Model**: Trillsson v1 achieves **AUC 0.98** with only **5.5M parameters**.

3. **Best Classifier**: Random Forest outperforms neural networks on foundation model embeddings.

4. **Key Innovation**: Unconstrained, open-vocabulary speech ("How are you feeling today?") is sufficient for high-performance PD detection.

5. **Model Ranking**: HuBERT > Trillsson > Wav2Vec2 > WavLM.

6. **Efficiency Insight**: Distilled models (Trillsson) offer excellent performance with minimal parameters.

7.2 Contributions to the Field

Contribution	Significance
New US English Dataset	Fills geographic gap; clinically validated
Comprehensive Foundation Model Evaluation	12 variants, 5 classifiers
Unconstrained Speech Validation	Real-world feasibility demonstrated
Efficiency-Accuracy Tradeoff Analysis	Identified optimal models for different scenarios
Practical Deployment Insights	No fine-tuning needed; simple classifiers work

7.3 Final Remarks

This study demonstrates that off-the-shelf speech foundation models, particularly HuBERT Large and Trillsson, combined with simple classifiers like Random Forest, achieve excellent PD detection performance (UAR 0.97) on unconstrained speech in a real-world clinical setting. The approach requires no fine-tuning, making it practical for deployment. While challenges remain—particularly regarding demographic biases, external validation, and interpretability—this work provides strong evidence for the viability of foundation model-based vocal biomarker analysis in clinical practice.

The focus on unconstrained speech is particularly significant: patients can simply answer "How are you feeling today?" during routine clinical visits, enabling passive, non-invasive screening without special equipment or training. This represents a substantial step toward scalable, accessible PD screening that could improve early detection and outcomes.

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